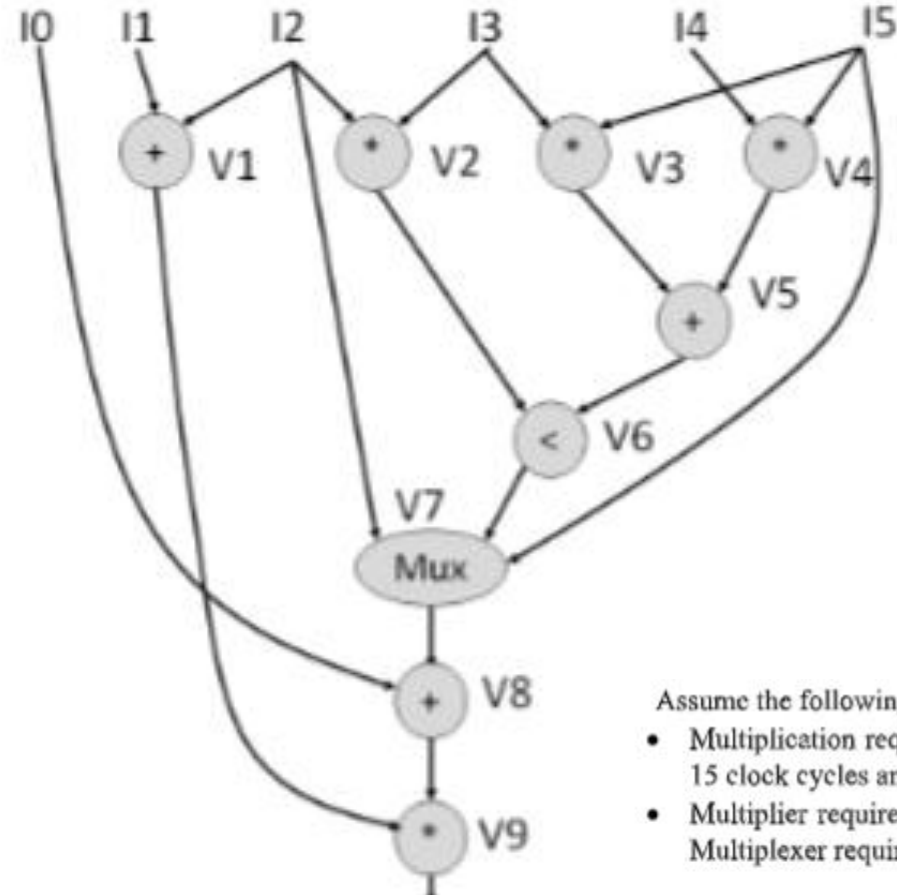


Revision-2

Consider the data-flow graph shown below:



Assume the following parameter values for FPGA implementation:

- Multiplication requires 50 clock cycles, Addition requires 20 clock cycles, comparison requires 15 clock cycles and the multiplexing operation requires 10 clock cycles.
- Multiplier requires 100 LUTs, Adder requires 20 LUTs, comparator requires 10 LUTs and the Multiplexer requires 8 LUTs.

- Draw the sequencing graph for the above data-flow graph.
- For an FPGA with 150 LUTs, apply list scheduling based temporal partitioning algorithm for partitioning the sequencing graph in (a). (Assume priority is assigned based on number of successors) (Clearly show which nodes are mapped to which partition)
- Calculate the quality of the partitioning obtained in (b).
- Calculate the wasted resources for the partitioning obtained in (b), by clearly showing the graph of LUT utilization with respect to run time for each partition (For each node in a partition, mark LUT usage on X-axis and runtime on Y-axis).

While designing a new PE (programmable element) structure, designers have to make a choice between using a small size LUT with additional elements or a large size LUT. Assume that your manager has asked you to design a combinational 3-input, 3-output PE structure, which is optimal for implementation of a 4-bit incrementer circuit (Assume $A(A_3A_2A_1A_0)$ as input and $O(O_3O_2O_1O_0)$ as output. If input is 0000, incrementer output will be 0001)

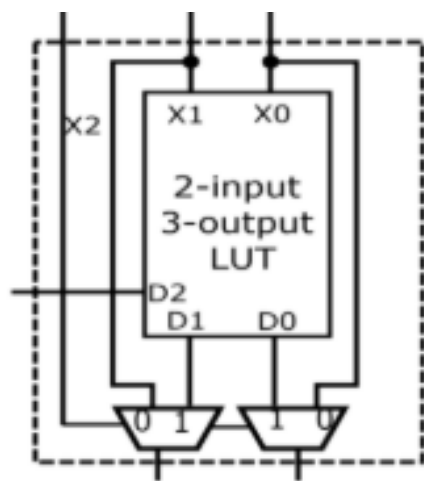
Initially, you have come up with two different designs. Figure (a) shows first design with one 2-input, 3-output LUT and two 2:1 multiplexers. Figure (b) shows second design with just one 3-input, 3-output LUT and no additional elements.

To determine which PE design is better, implement a 4-bit incrementer optimally using

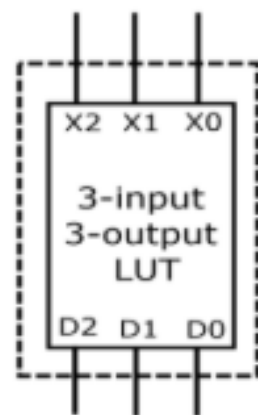
(a) least number of PEs, when PE structure is as shown in Figure (a)

(b) least number of PEs, when PE structure is as shown in Figure (b)

Comment on advantages and disadvantages of both the PE designs with respect to the implementation of 4-bit incrementer.



(a): PE Design 1



(b): PE Design 2